

Greedy Algorithm

① Optimal solⁿ (max, min)
↳ Sorting ✓

② No fixed rule

③ lots of time global optimum solⁿ is not achieved.

Greedy Algo is the problem solving technique where we make the locally optimal choices ^(best) at each stage and try to achieve global optimum solⁿ.

Question:

Min Absolute difference pair:

Given 2 array A and B of same size. pair $(i, j) \mid i \in A, j \in A$

Sum = $\sum |i - j|$, find min. sum.

A = [1, 2, 3]

B = [2, 1, 3]

↳ combinations

Can 1 (1, 2) (2, 1) (3, 3) → Sum = 1 + 1 + 0 = 2

Can 2 (1, 1) (2, 2) (3, 3) → Sum = 0 + 0 + 0 = 0 (min)

Can 3 (1, 3) (2, 1) (3, 1) → Sum = 2 + 1 + 1 = 4

⋮

Sum = 0

for (i = 0 to n) {

Sum += Math.abs(A[i] - B[i]);

}

ans Sum

A → Sort [1, 2, 3]

B → Sort [1, 2, 3]

